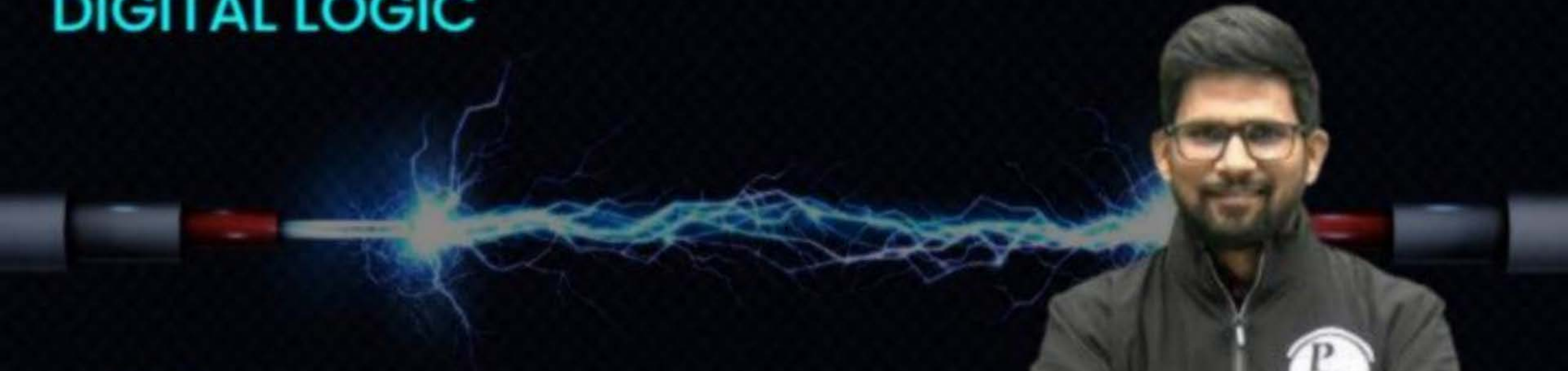
# COMPUTER SCIENCE & IT

## DIGITAL LOGIC



Lecture No. 07

## BOOLEAN THEOREMS AND GATES



By- Chandan Gupta Sir




XOR & XNOR GATE




- XOR & XNOR GATE Cont.
- Question Discussion

# HW

#

1. $\overline{AB \oplus BC} = ? \longrightarrow \overline{B(A \oplus C)} = \overline{B} + \overline{A \oplus C} = \overline{B} + A \odot C = AB \odot BC$

2. $\overline{A \oplus \overline{B} \oplus \overline{C} \oplus \overline{D}} = ? \Rightarrow A \oplus B \oplus C \oplus D = \overline{A \odot B \odot C \odot D}$

3. $\underline{AB} \oplus \underline{\overline{B}C} = ? \Rightarrow (AB + \overline{B}C)$

$$\underline{AB} \oplus \underline{\overline{B}C} = \overline{AB} \cdot \overline{\overline{B}C} + AB \cdot \overline{\overline{B}C}$$

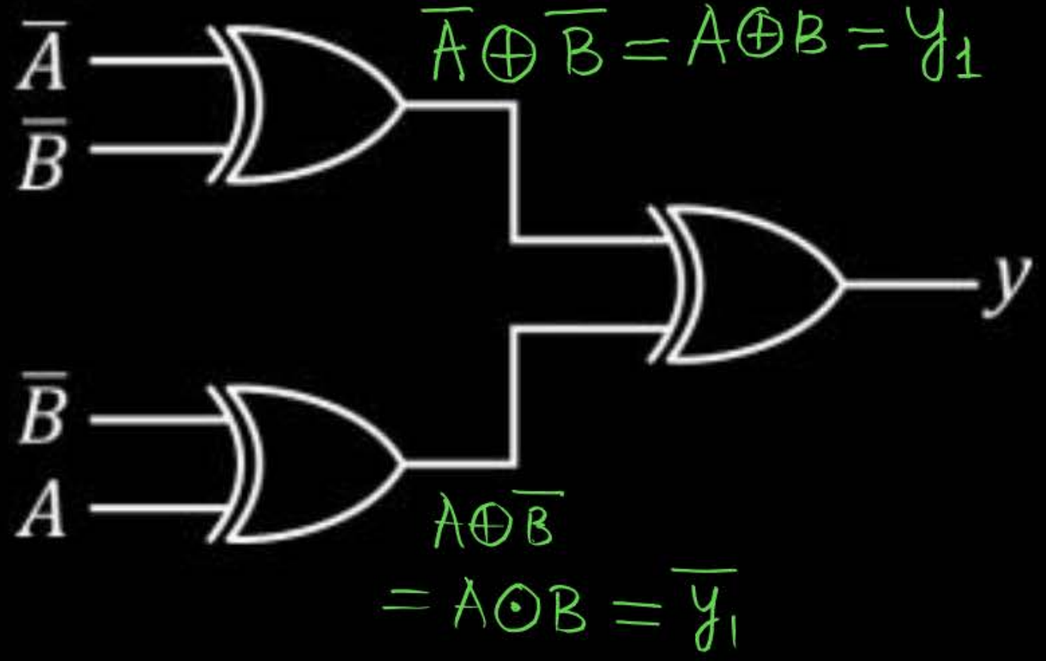
4. $AB \oplus \overline{BC} = ? \Rightarrow \overline{AB \oplus BC} = \overline{B(A \oplus C)} = \overline{B} + A \odot C$

5. $\underline{\overline{A}B} \oplus \underline{A\overline{B}} = ? \Rightarrow \overline{A}B + A\overline{B} = A \oplus B$

6. $\underline{\overline{A}\,\overline{B}} \oplus \underline{AB} = ? \Rightarrow \overline{A}\,\overline{B} + AB = A \odot B$

$$\overline{(A+B)} \oplus AB = \overline{(A+B) \oplus AB}$$

## Question




$\overline{A} \oplus \overline{B} = A \oplus B = y_1$

$y = y_1 \oplus \overline{y_1}$

$= 1$

$A \oplus \overline{B}$

$= A \odot B = \overline{y_1}$

Output $y$ is

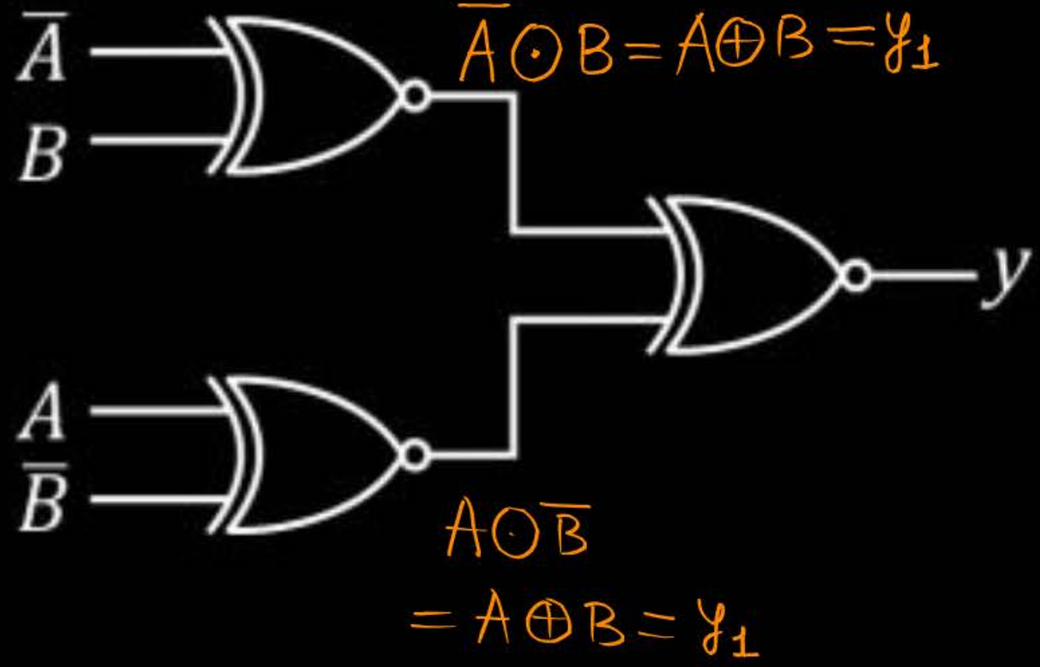
(a) $A \odot B$

(b) $A \oplus B$

(c) Always '1'

(d) Always '0'

## Question




Output $y$ is

(a) $A \oplus B$

(b) $A \odot B$

(c) Always '1'

(d) Always '0'

# Question

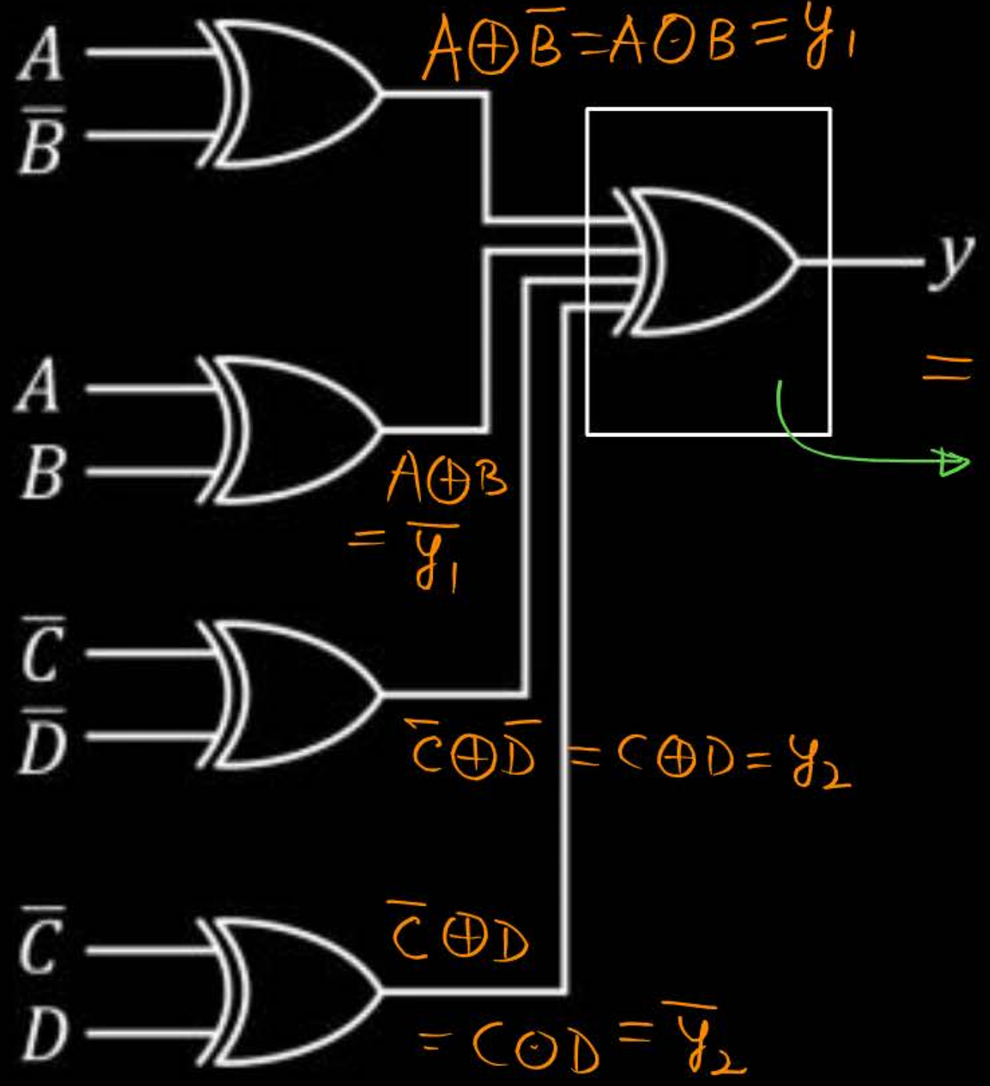
Output $y$ is

(a) $A \odot B \oplus C \oplus D$

(b) $A \oplus B \oplus C \oplus D$

(c) Always '0'

(d) $A \odot B \odot C \odot D$



## Question

$y = \overline{\overline{A \oplus B} \oplus \bar{C}}$ will be equal to $= \overline{A \oplus B \oplus C} = A \odot B \oplus C = A \oplus B \odot C$

(a) $A \odot B \oplus C =$

(b) $y = \overline{\bar{A} \oplus \overline{B \odot C}} = \overline{A \oplus B \odot C}$

(c) $y = \overline{A \oplus B \oplus \bar{C}}$

$A \oplus B \oplus C$

(d) None of these

## Question

$y = A \oplus [A + B]$ equal to

(a) $A \oplus B$ ✗

(b) $y = A \odot B$

(c) $\bar{A}B$

(d) $A\bar{B}$

Note: XOR & XNOR does not hold distributive law over 'OR' & over 'AND' operation.

$$A \oplus (A+B) \neq A \oplus A + A \oplus B = 0 + A \oplus B = A \oplus B$$

$$A \oplus (A+B) = \bar{A} \cdot (A+B) + A \cdot \overline{(A+B)} = \bar{A}B + A \cdot \bar{A} \cdot \bar{B} = \bar{A}B$$

$$A \oplus (A \cdot B) \neq (A \oplus A) \cdot (A \oplus B)$$

## Question

$y = A \oplus B \oplus AB$ equal to

(a) $AB$ (b) $(A + B)$

(c) $\overline{AB}$ (d) $\overline{A + B}$

$A \oplus B \oplus AB = A \oplus [B \oplus AB] = A \oplus [B(1 \oplus A)] = A \oplus [B \cdot \bar{A}]$

$= \bar{A} \cdot \overline{\bar{A} \cdot B} + A \cdot \overline{\overline{\bar{A} \cdot B}}$

$= \bar{A} \cdot B + A \cdot (A + \bar{B}) = \bar{A}B + A = A + (\bar{A} \cdot B) = (A + \bar{A})(A + B)$

$= (A + B)$

$A \oplus \bar{A} \cdot B$

$= \overline{\bar{A} \oplus \bar{A} \cdot B}$

$= \overline{\bar{A}[1 \oplus B]}$

$= \overline{\bar{A} \cdot \bar{B}} = (A + B)$

$$A \oplus B \oplus AB = [A \oplus B] \oplus AB$$

$$\overline{\bar{A}B + A\bar{B}}$$

$$= (\bar{A}B + A\bar{B}) \oplus AB$$

$$= (\bar{A}\bar{B} + AB) \cdot AB + (\bar{A}B + A\bar{B}) \cdot \overline{AB}$$

$$= AB + (\bar{A}B + A\bar{B})(\bar{A} + \bar{B})$$

$$= AB + \bar{A}B + A\bar{B}$$

$$= B(A + \bar{A}) + A\bar{B}$$

$$= B + (A\bar{B}) = (B + A) \cdot (B + \bar{B}) = B + A = (A + B)$$

- $\bar{A} \oplus B \oplus AB = (A+B)$

$A \oplus B \oplus (A+B) = \bar{A}B$

- $A \oplus B \oplus \bar{A} \cdot B = \overline{\bar{A} \oplus B \oplus \bar{A} \cdot B}$

$= \overline{(\bar{A}+B)}$

$= A \cdot \bar{B}$

- $A \oplus B \oplus \bar{A} \cdot \bar{B} = \bar{A} \oplus \bar{B} \oplus \bar{A} \cdot \bar{B} = (\bar{A}+\bar{B})$

$\overline{A \oplus B \oplus C} = \bar{A} \oplus B \oplus C$

$= A \oplus \bar{B} \oplus C$

$= A \oplus B \oplus \bar{C}$

$A \oplus B \oplus \overline{(A+B)} = \overline{A \oplus B \oplus (A+B)}$

$= \overline{\bar{A} \cdot \bar{B}} = (\bar{A}+\bar{B})$

- $A \oplus \bar{B} \oplus \bar{A} \cdot B = \bar{A} \oplus B \oplus \bar{A} \cdot B = (\bar{A}+B)$

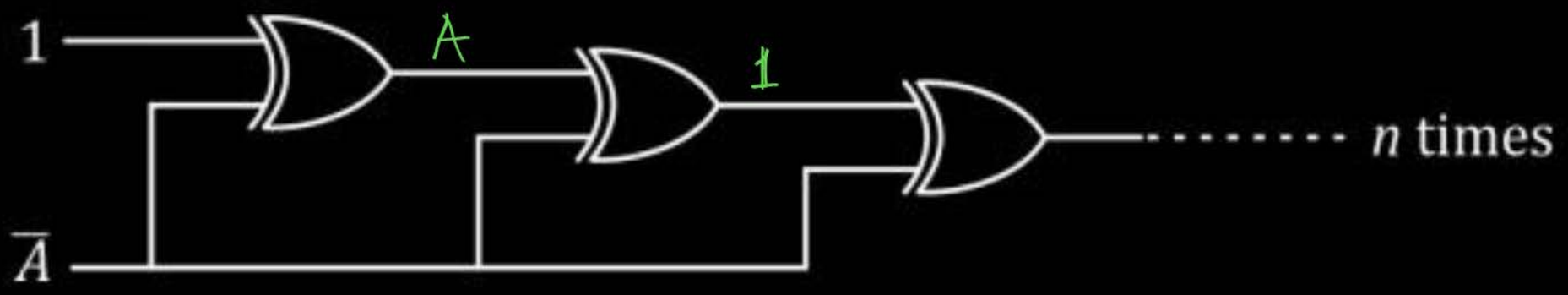
- $\bar{A} \oplus \bar{B} \oplus (\bar{A}+B)$

$= \overline{\bar{A} \oplus B} \oplus (\bar{A}+B)$

$= \bar{A} \cdot B = (A+\bar{B})$

$= A \oplus B \oplus \overline{\bar{A} \cdot B}$

## Question




Then which of the following is true :

(a) Output is $A$ if $n$ is even

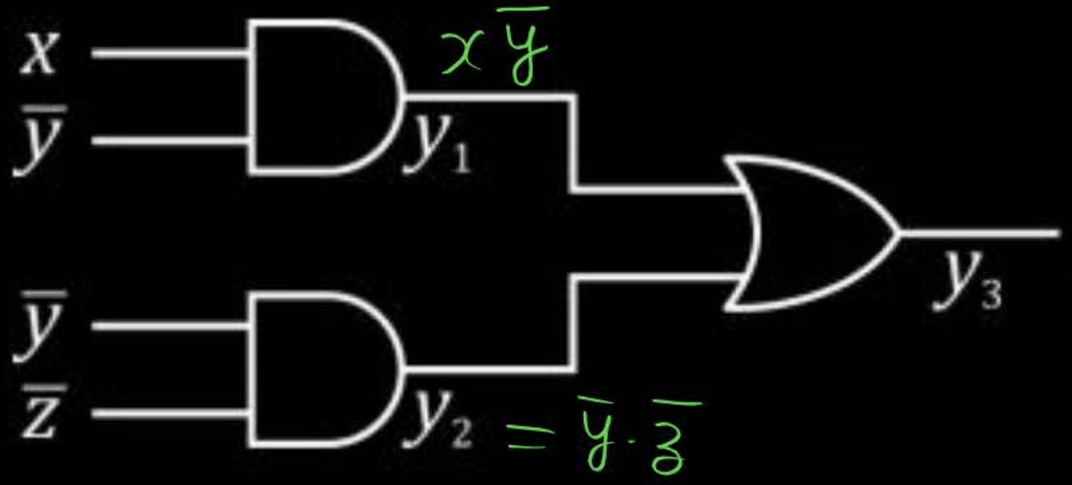
(b) Output is $\bar{A}$ if $n$ is even

(c) Output is $A$ if $n$ is odd

(d) Output is 1 if $n$ is odd

## Question

A digital circuit is implemented as :




$x\bar{y}$

$y_2 = \bar{y}\cdot\bar{z}$

$y_3 = x(\bar{y} + \bar{y}\bar{z} = y_1 + y_2$

Then output $y = y_1 \oplus y_2 \oplus y_3$ will be

$= y \Rightarrow y_1 \oplus y_2 \oplus y_3 = y_1 \oplus y_2 \oplus (y_1 + y_2)$

$= y_1 \cdot y_2$

$= x\bar{y}\cdot\bar{y}\bar{z}$

$= x\bar{y}\cdot\bar{z}$

(a) $x\bar{y} + \bar{y}\bar{z}$

(b) $x \oplus y \oplus z$

(c) $x\bar{y}\bar{z}$

(d) $x + \bar{y} + \bar{z}$

- $f(A,B,C) = AB + BC \Rightarrow AB(\bar{C}+C) + (\bar{A}+A)BC$

$= AB\bar{C} + ABC + \bar{A}BC + ABC = \underset{110}{AB\bar{C}} + \underset{111}{ABC} + \underset{011}{\bar{A}BC}$

$= \Sigma(3,6,7)$

$f_1(A,B,C) = A\bar{B} + \bar{B}C + AB\bar{C}$

$= A\bar{B}(\bar{C}+C) + (\bar{A}+A)\bar{B}C + AB\bar{C}$

$= A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C + A\bar{B}C + AB\bar{C}$

$= \underset{100}{A\bar{B}\bar{C}} + \underset{101}{A\bar{B}C} + \underset{001}{\bar{A}\bar{B}C} + \underset{110}{AB\bar{C}}$

$= \Sigma(1,4,5,6)$

$$f(A,B,C,D) = BD + \bar{A}C + AB\bar{C}$$

$$= (\bar{A}+A)\,B\,(\bar{C}+C)\,D + \bar{A}(\bar{B}+B)\,C\,(\bar{D}+D) + AB\bar{C}(\bar{D}+D)$$

$$= \underbrace{(\bar{A}\bar{C}+\bar{A}C+A\bar{C}+AC)}_{5,\ 7}\underline{\underline{BD}} + (\bar{B}\bar{D}+\bar{B}D+B\bar{D}+BD)\,\bar{A}C + AB\bar{C}\,\bar{D} + AB\bar{C}D$$

$$= \underset{0\,1\,0\,1}{\bar{A}B\bar{C}D} + \underset{0\,1\,1\,1}{\bar{A}BCD} + \underset{1\,1\,0\,1}{AB\bar{C}D} + \underset{1\,1\,1\,1}{ABCD} \quad (13,\ 15)$$

$$= \underset{0\,0\,1\,0}{\bar{A}\bar{B}C\bar{D}} + \underset{0\,0\,1\,1}{\bar{A}\bar{B}CD} + \underset{0\,1\,1\,0}{\bar{A}BC\bar{D}} + \underset{0\,1\,1\,1}{\bar{A}BCD} + \underset{1\,1\,0\,0}{AB\bar{C}\bar{D}} + \underset{1\,1\,0\,1}{AB\bar{C}D}$$

$$(2) \quad (3) \quad (6) \quad (7) \quad 12 \quad 13$$

$$= \Sigma(2,3,5,6,7,12,13,15)$$

- $f(A,B,C) = AB + BC = \Sigma(3,6,7)$

AB

1 1 0 → 6

1 1 1 → 7

BC

0 1 1 → 3

1 1 1 → 7

$f(A,B,C,D) = BD + \bar{A}C + AB\bar{C}$

B D

0 1 0 1 → 5

0 1 1 1 → 7

1 1 0 1 → 13

1 1 1 1 → 15

$\bar{A}$ C

0 0 1 0 → 2

0 0 1 1 → 3

0 1 1 0 → 6

0 1 1 1 → 7

AB$\bar{C}$

1 1 0 0 → 12

1 1 0 1 → 13

$= \Sigma(2,3,5,6,7,12,13,15)$

$f(A,B,C,D) = AD + B\bar{C} + A\bar{B}\bar{D} = \Sigma(4,5,8,9,10,11,12,13,15)$

| A | D | |
|---|---|---|
| 1 | 0 0 1 | = 9 |
| 1 | 0 1 1 | = 11 |
| 1 | 1 0 1 | = 13 |
| 1 | 1 1 1 | = 15 |

| B | $\bar{C}$ | | |
|---|---|---|---|
| 0 | 1 | 0 | 0 = 4 |
| 0 | 1 | 0 | 1 = 5 |
| 1 | 1 | 0 | 0 = 12 |
| 1 | 1 | 0 | 1 = 13 |

| A | $\bar{B}$ | $\bar{D}$ | |
|---|---|---|---|
| 1 | 0 | 0 | 0 = 8 |
| 1 | 0 | 1 | 0 = 10 |

# H.W.

Q.1 $A\bar{B} \oplus \bar{B}C \oplus (\bar{A} + B + \bar{C})$ is equal to

a. $B + \bar{A}\bar{C}$ b. $\bar{B} + AC$ c. $AB\bar{C}$ d. $A\bar{B}C$

Q.2 $A \oplus (\bar{A} + \bar{B})$ is equal to

a. $A\bar{B}$ b. $\bar{A}B$ c. $\bar{A} + B$ d. $A + \bar{B}$

Q.3 $\bar{A} \oplus \bar{B} \oplus \bar{A}\bar{B} \oplus ABC$ is equal

a. $\bar{A} + \bar{B} + \bar{C}$ b. $A + B + C$ c. $\bar{A} + \bar{B} + C$ d. $(AB + BC)$

Q.4 $(A \oplus \bar{B}) \oplus (C \oplus \bar{B}) \oplus (B \oplus \bar{A}) \oplus (\bar{A} \oplus \bar{C})$ is equal to

a. $A \oplus B \oplus C$ b. $A \odot B \oplus C$ c. $(A \odot B)$ d. $(A \oplus B)$

Q.5 $f(A, B, C, D)$

$= AC\bar{D} + \bar{B}\bar{D} + \bar{A}\bar{C} + ABC$

$= \Sigma(\ ?\ )$

Q.6 $f(A, B, C)$

$= A + \bar{B}\bar{C} + \bar{A}C$

$= \Sigma(\ ?\ )$

Q.7 $f(A,B,C) = \underline{\underline{(B+\bar{C})}}\,(\bar{A}+\bar{B})(A+\bar{B}+C) = \Pi(?) = \Sigma(?)$

| | B | $\bar{C}$ | |
|---|---|---|---|
| 0 | 0 | $1 \rightarrow 1$ | } POS |
| 1 | 0 | $1 \rightarrow 5$ | } POS |



# 2 Minute Summary

XOR & XNOR gate

Thank you

GW

*Soldiers !*

